

Dynamical Modelling & Control of an Underactuated Articulated-Soft Robotic Manipulator

7CCEMRDC Robot Dynamics & Control — Individual Coursework

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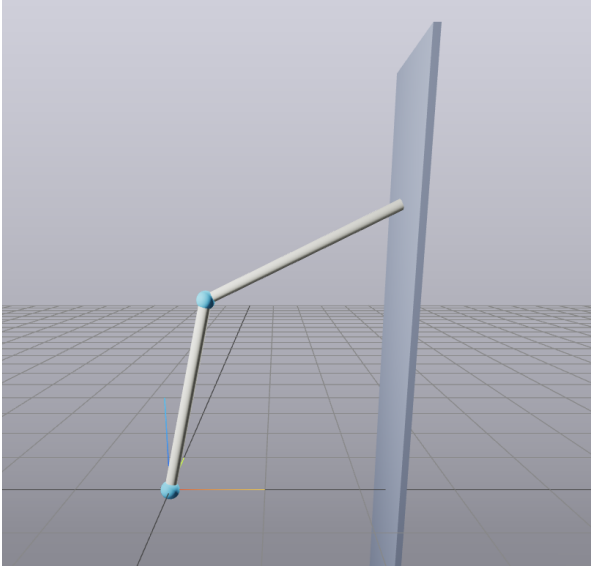
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1 System Description and Dynamical Model

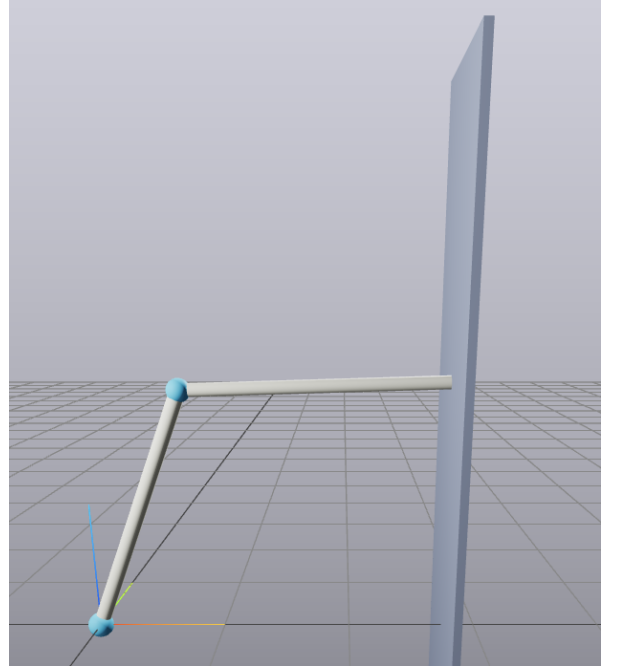
1.1 System overview

The system under consideration is a two-degree-of-freedom (2-DoF) planar *articulated-soft robot* (ASR) manipulator operating in the xz -plane under gravity. Each joint consists of a motor (rotor) connected to its corresponding rigid link through a passive rotational spring, making the system underactuated: the input vector $\boldsymbol{\tau}_m \in \mathbb{R}^2$ drives four generalised coordinates $\mathbf{q}_T = [\mathbf{q}^\top, \boldsymbol{\theta}^\top]^\top \in \mathbb{R}^4$, where $\mathbf{q} = [q_1, q_2]^\top$ are the link angles and $\boldsymbol{\theta} = [\theta_1, \theta_2]^\top$ are the motor angles.

The robot is tasked with pressing its end-effector (EE) against a moving wall located at $x \approx 1.085$ m and drawing a vertical sinusoidal line by oscillating the EE in the z -direction, thereby combining trajectory tracking with compliant physical interaction.



(a) Initial configuration: EE approaching the wall.



(b) Final configuration: EE in sustained wall contact at lower drawing position.

Figure 1: MeshCat visualisations of the ASR manipulator during the wall-drawing task. The two cyan spheres are the motor rotors; the rigid links are shown in light grey. The wall (blue-grey panel) is constrained to slide only in the x -direction.

1.2 Flexible-joint dynamics

Following the Spong flexible-joint model [1, 7], the full equations of motion are obtained by coupling the link-side Newton–Euler equations with motor-side dynamics via the joint spring torques:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{K}(\mathbf{q} - \boldsymbol{\theta}) + \boldsymbol{\tau}_g(\mathbf{q}) = \mathbf{0} \quad (1)$$

$$\mathbf{J}\ddot{\boldsymbol{\theta}} + \mathbf{D}\dot{\boldsymbol{\theta}} + \mathbf{K}(\boldsymbol{\theta} - \mathbf{q}) = \boldsymbol{\tau}_m \quad (2)$$

where $\mathbf{K} = \text{diag}(k_1, k_2)$ is the joint stiffness matrix and $\boldsymbol{\delta} = \mathbf{q} - \boldsymbol{\theta}$ is the spring deflection vector. The link-side dynamics (eq. (1)) contain no direct motor input: the springs couple the two subsystems, reflecting the underactuated nature of the ASR.

1.3 Two-link planar manipulator matrices

Modelling link masses as point masses at the link endpoints, the standard Lagrangian derivation yields:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} (m_1 + m_2)L_1^2 + m_2L_2^2 + 2m_2L_1L_2 \cos q_2 & m_2L_2^2 + m_2L_1L_2 \cos q_2 \\ m_2L_2^2 + m_2L_1L_2 \cos q_2 & m_2L_2^2 \end{bmatrix} \quad (3)$$

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -2h\dot{q}_2 & -h\dot{q}_2 \\ h\dot{q}_1 & 0 \end{bmatrix}, \quad h = m_2L_1L_2 \sin q_2 \quad (4)$$

$$\boldsymbol{\tau}_g(\mathbf{q}) = \begin{bmatrix} (m_1 + m_2)gL_1 \sin q_1 + m_2gL_2 \sin(q_1 + q_2) \\ m_2gL_2 \sin(q_1 + q_2) \end{bmatrix} \quad (5)$$

Substituting the numerical parameters (see table 1) gives

$$M_{11} = 4.5 + 3 \cos q_2, \quad M_{12} = M_{21} = 1.5 + 1.5 \cos q_2, \quad M_{22} = 1.5 \quad (6)$$

The inertia matrix $\mathbf{M}(\mathbf{q})$ is symmetric positive definite for all $q_2 \notin \{0, \pi\}$, which corresponds to the singular (fully extended or fully folded) configurations.

1.4 Motor inertia

The motor rotors are modelled as solid spheres with inertia $J_i = \frac{2}{5}m_{r_i}r_i^2$. The motor inertia matrix is diagonal: $\mathbf{J} = \text{diag}(J_1, J_2)$.

1.5 Simulation parameters

Table 1: Simulation model parameters.

Parameter	Symbol	Value	Unit
Link lengths	$L_1 = L_2$	1.0	m
Link masses	$m_1 = m_2$	1.5	kg
Rotor 1 mass	m_{r_1}	0.5	kg
Rotor 2 mass	m_{r_2}	0.4	kg
Rotor radius	r	0.08	m
Motor inertia 1	J_1	0.00128	kg m ²
Motor inertia 2	J_2	0.001024	kg m ²
Spring stiffness 1	k_1	30.5	N m/rad
Spring stiffness 2	k_2	10.01	N m/rad
Motor damping	d	0.1	N m s/rad
Gravity	g	9.81	m/s ²
Torque limit	$\tau_{\max}$	200	N m

2 Control Design Rationale

2.1 Singular perturbation reduction

The full 4-DoF flexible-joint model (eqs. (1) and (2)) is difficult to control directly. Singular perturbation theory [1, 8] states that in the limit $\mathbf{K} \rightarrow \infty$ — i.e. when the springs are stiff relative to the dynamics

— the deflection $\delta \rightarrow \mathbf{0}$ and the two subsystems collapse to a single rigid-body equation augmented with motor inertia:

$$(\mathbf{M}(\mathbf{q}) + \mathbf{J})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \tau_g(\mathbf{q}) = \tau_m \quad (7)$$

This approximation is valid for the present system because the spring stiffnesses ($k_1 = 30.5$, $k_2 = 10.01$ N m/rad) are large relative to the gravitational loading at operating configurations and the spring deflections observed in simulation remain small (see section 5). All subsequent controller design uses eq. (7); the full flexible-joint simulator (eqs. (1) and (2)) is used for verification.

2.2 Computed torque control law

Computed torque control (CTC) achieves exact linearisation by inverting the plant nonlinearities [2, 4]. Define the joint-space tracking error $\mathbf{e} = \mathbf{q}_{\text{des}} - \mathbf{q}$ and introduce the auxiliary input $\mathbf{v}$:

$$\tau_m = (\mathbf{M}(\mathbf{q}) + \mathbf{J})\mathbf{v} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \tau_g(\mathbf{q}) \quad (8)$$

where

$$\mathbf{v} = \ddot{\mathbf{q}}_{\text{des}} + K_d(\dot{\mathbf{q}}_{\text{des}} - \dot{\mathbf{q}}) + K_p(\mathbf{q}_{\text{des}} - \mathbf{q}) \quad (9)$$

Substituting eq. (8) into eq. (7) yields the decoupled, linear error dynamics

$$\ddot{\mathbf{e}} + K_d\dot{\mathbf{e}} + K_p\mathbf{e} = \mathbf{0} \quad (10)$$

The gain matrices are chosen as $K_p = \text{diag}(100, 100)$ rad⁻² and $K_d = \text{diag}(20, 20)$ rad⁻¹s, yielding per-joint characteristic equations

$$s^2 + 20s + 100 = (s + 10)^2 = 0 \quad (11)$$

i.e. a repeated real eigenvalue at $s = -10$, corresponding to natural frequency $\omega_n = 10$ rad/s and damping ratio $\zeta = 1$ (critically damped). Critical damping eliminates overshoot while maximising convergence speed.

2.3 Desired trajectory in joint space

To track a Cartesian trajectory $\mathbf{p}_{\text{des}}(t)$, the desired joint states are computed at each control step:

1. **Desired angles:** $\mathbf{q}_{\text{des}} = \text{IK}(\mathbf{p}_{\text{des}})$ (see section 3).
2. **Desired velocities:** $\dot{\mathbf{q}}_{\text{des}} = \mathbf{J}(\mathbf{q}_{\text{des}})^{-1}\dot{\mathbf{p}}_{\text{des}}$.
3. **Desired accelerations:** $\ddot{\mathbf{q}}_{\text{des}} = \mathbf{J}(\mathbf{q}_{\text{des}})^{-1}(\ddot{\mathbf{p}}_{\text{des}} - \dot{\mathbf{J}}\dot{\mathbf{q}}_{\text{des}})$.

2.4 Link-angle computation from sensor feedback

The Drake simulator provides motor angles $\boldsymbol{\theta}$ and spring deflections $\boldsymbol{\delta}$ directly. Link angles and velocities are recovered as

$$\mathbf{q} = \boldsymbol{\theta} + \boldsymbol{\delta}, \quad \dot{\mathbf{q}} = \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\delta}} \quad (12)$$

2.5 Desired spring deflection and motor angle

To facilitate analysis and logging, the expected spring deflection at the desired configuration is computed from the link-side equilibrium condition (treating link-side acceleration as zero in the steady-state):

$$\mathbf{K}\delta_{\text{des}} = -[\mathbf{M}(\mathbf{q}_{\text{des}})\ddot{\mathbf{q}}_{\text{des}} + \mathbf{C}(\mathbf{q}_{\text{des}}, \dot{\mathbf{q}}_{\text{des}})\dot{\mathbf{q}}_{\text{des}} + \boldsymbol{\tau}_g(\mathbf{q}_{\text{des}})] \quad (13)$$

The desired motor angles are then $\boldsymbol{\theta}_{\text{des}} = \mathbf{q}_{\text{des}} - \delta_{\text{des}}$.

2.6 Interaction control via inherent compliance

The spring elements provide natural impedance at the joints. When the EE contacts the wall, the spring deflection absorbs the contact force: $F_{\text{contact}} \approx k_i \delta_i / L_i$. The trajectory is commanded 3 mm past the nominal wall surface ($x_{\text{des}} = 1.088$ m vs. wall surface at approximately 1.085 m), ensuring sustained contact without excessive wall penetration. The large wall damping (180 Ns/m + 350 Ns/m extra) and mass (150 kg) ensure the wall remains nearly stationary; the contact force is therefore absorbed elastically by the spring joints rather than displacing the wall. This constitutes an implicit compliance strategy analogous to series elastic actuation.

2.7 Control system architecture

Figure 2 summarises the complete closed-loop architecture implemented in `Coursework_student.py`.

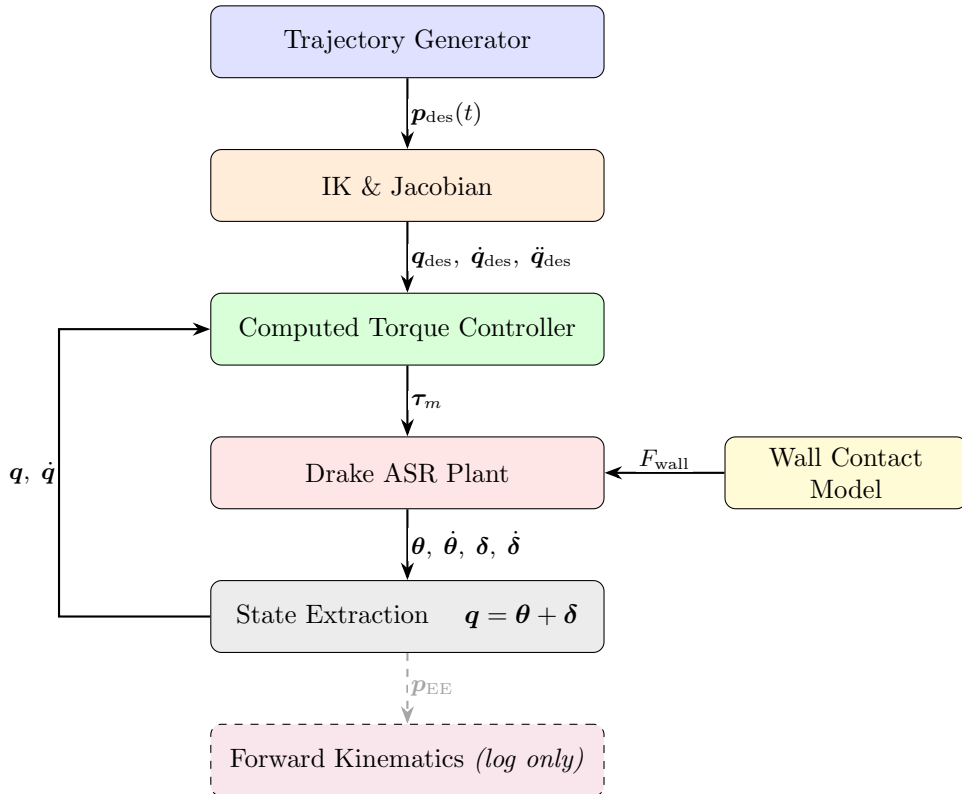


Figure 2: Closed-loop control architecture implemented in `Coursework_student.py`. The main path (solid arrows, centre column) runs top to bottom: the trajectory generator outputs $\mathbf{p}_{\text{des}}(t)$; IK and Jacobian inversion produce joint-space references; the computed-torque controller (eq. (8)) outputs motor torques $\boldsymbol{\tau}_m$; the Drake plant integrates the full flexible-joint dynamics (eqs. (1) and (2)); state extraction recovers link angles $\mathbf{q} = \boldsymbol{\theta} + \boldsymbol{\delta}$ and velocities. The feedback path (solid arrow, left) returns $\mathbf{q}, \dot{\mathbf{q}}$ to the controller each timestep. The wall contact model (right) runs in parallel, injecting F_{wall} into the plant’s actuation port. The dashed arrow is a logging-only path for post-simulation plotting.

3 Forward Kinematics, Inverse Kinematics, and the Jacobian

3.1 Forward kinematics

For a 2-DoF planar manipulator with link lengths $L_1 = L_2 = 1.0$ m [5], the end-effector position in the xz -plane is

$$\mathbf{p} = \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} L_1 \cos q_1 + L_2 \cos(q_1 + q_2) \\ L_1 \sin q_1 + L_2 \sin(q_1 + q_2) \end{bmatrix} \quad (14)$$

3.2 Inverse kinematics

Given a desired EE position (x_d, z_d) , the elbow-up solution is obtained via the law of cosines:

$$D = \frac{x_d^2 + z_d^2 - L_1^2 - L_2^2}{2L_1L_2} \in [-1, 1] \quad (15)$$

$$q_2^* = \arccos D \quad (16)$$

$$q_1^* = \arctan\left(\frac{z_d}{x_d}\right) - \arctan\left(\frac{L_2 \sin q_2^*}{L_1 + L_2 \cos q_2^*}\right) \quad (17)$$

The argument D is clipped to $[-1, 1]$ to guard against numerical rounding outside the reachable workspace. The elbow-up branch ($q_2 > 0$) is selected throughout, consistent with the robot's initial configuration and the wall-side workspace.

3.3 Geometric Jacobian

Differentiating eq. (14) with respect to $\mathbf{q}$:

$$\mathbf{J}(\mathbf{q}) = \frac{\partial \mathbf{p}}{\partial \mathbf{q}} = \begin{bmatrix} -L_1 \sin q_1 - L_2 \sin(q_1 + q_2) & -L_2 \sin(q_1 + q_2) \\ L_1 \cos q_1 + L_2 \cos(q_1 + q_2) & L_2 \cos(q_1 + q_2) \end{bmatrix} \quad (18)$$

The Jacobian relates Cartesian and joint-space velocities: $\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$.

3.4 Jacobian time derivative

The term $\dot{\mathbf{J}}\dot{\mathbf{q}}$ is required for the feedforward acceleration. Denoting $q_{12} = q_1 + q_2$:

$$\dot{\mathbf{J}}\dot{\mathbf{q}} = \begin{bmatrix} (-L_1 \cos q_1 \dot{q}_1 - L_2 \cos q_{12}(\dot{q}_1 + \dot{q}_2))\dot{q}_1 + (-L_2 \cos q_{12}(\dot{q}_1 + \dot{q}_2))\dot{q}_2 \\ (-L_1 \sin q_1 \dot{q}_1 - L_2 \sin q_{12}(\dot{q}_1 + \dot{q}_2))\dot{q}_1 + (-L_2 \sin q_{12}(\dot{q}_1 + \dot{q}_2))\dot{q}_2 \end{bmatrix} \quad (19)$$

3.5 Desired Cartesian trajectory

The robot is commanded to draw a vertical sinusoidal line at constant x -depth against the wall while oscillating vertically:

$$x_{\text{des}}(t) = 1.088 \text{ m} \quad (20)$$

$$z_{\text{des}}(t) = 0.40 + 0.40 \sin(2\pi f_d t), \quad f_d = 0.5 \text{ Hz} \quad (21)$$

The corresponding velocity and acceleration feedforward signals are

$$\dot{z}_{\text{des}}(t) = 0.40 \cdot 2\pi f_d \cos(2\pi f_d t) = 0.4\pi \cos(\pi t) \text{ m/s} \quad (22)$$

$$\ddot{z}_{\text{des}}(t) = -0.40(2\pi f_d)^2 \sin(2\pi f_d t) = -0.4\pi^2 \sin(\pi t) \text{ m/s}^2 \quad (23)$$

The desired x -velocity and acceleration are both zero, so the Jacobian inversion primarily maps the z -components. The EE is commanded to operate in the range $z \in [0.0, 0.8] \text{ m}$, well within the reachable workspace.

4 Closed-Loop Stability Analysis

4.1 Eigenvalue analysis

Assuming perfect model knowledge, substituting the CTC law (eqs. (8) and (9)) into eq. (7) yields the decoupled error dynamics eq. (10). Since K_p and K_d are diagonal, the two joints decouple into identical second-order systems. The characteristic polynomial for each joint is

$$p(s) = s^2 + K_{d,ii} s + K_{p,ii} = s^2 + 20s + 100 = (s + 10)^2 \quad (24)$$

The eigenvalues are $\lambda_{1,2} = -10$, both in the open left-half complex plane. The closed-loop system is therefore asymptotically stable with a critically-damped response: errors decay as $e(t) = (a + bt)e^{-10t}$, converging to zero without overshoot.

4.2 Lyapunov stability

A rigorous proof of global asymptotic stability (under perfect model knowledge) follows from the candidate Lyapunov function

$$V(\mathbf{e}, \dot{\mathbf{e}}) = \frac{1}{2} \dot{\mathbf{e}}^\top \dot{\mathbf{e}} + \frac{1}{2} \mathbf{e}^\top K_p \mathbf{e} > 0 \quad \forall (\mathbf{e}, \dot{\mathbf{e}}) \neq \mathbf{0} \quad (25)$$

V is positive definite since $K_p \succ 0$. Its time derivative along trajectories of eq. (10) is

$$\dot{V} = \dot{\mathbf{e}}^\top \ddot{\mathbf{e}} + \mathbf{e}^\top K_p \dot{\mathbf{e}} = \dot{\mathbf{e}}^\top (-K_d \dot{\mathbf{e}} - K_p \mathbf{e}) + \mathbf{e}^\top K_p \dot{\mathbf{e}} = -\dot{\mathbf{e}}^\top K_d \dot{\mathbf{e}} \quad (26)$$

Since $K_d \succ 0$, equation eq. (26) gives $\dot{V} \leq 0$. $\dot{V} = 0$ only when $\dot{\mathbf{e}} = \mathbf{0}$; on this set, eq. (10) reduces to $K_p \mathbf{e} = \mathbf{0}$, which implies $\mathbf{e} = \mathbf{0}$ since $K_p \succ 0$. By LaSalle's Invariance Principle, the largest invariant set contained in $\{\dot{V} = 0\}$ is $\{(\mathbf{e}, \dot{\mathbf{e}}) = \mathbf{0}\}$; hence $\mathbf{e}(t) \rightarrow \mathbf{0}$ and $\dot{\mathbf{e}}(t) \rightarrow \mathbf{0}$ asymptotically.

4.3 Remarks on the flexible-joint system

The Lyapunov proof above holds exactly for the reduced model eq. (7). For the full flexible-joint system (eqs. (1) and (2)) the spring terms introduce a bounded perturbation to the error dynamics:

$$\ddot{\mathbf{e}} + K_d \dot{\mathbf{e}} + K_p \mathbf{e} = (\mathbf{M} + \mathbf{J})^{-1} \mathbf{K} \delta \quad (27)$$

Since $\mathbf{M} + \mathbf{J}$ is uniformly bounded below and the spring deflections remain bounded in simulation (order 10^{-2} rad for k_1), the perturbation term is small. For sufficiently large k the system remains stable and tracking errors remain small, which is confirmed by the simulation results in section 5.

5 Simulation Results

5.1 End-effector trajectory tracking

The EE trajectory (fig. 3, top-left) shows tight tracking of the desired sinusoidal path once the initial transient (approximately 0.5–1 s) has decayed. The wall contact points (purple markers) trace a near-vertical line at $x \approx 1.085\text{--}1.088\text{ m}$, demonstrating successful contact maintenance throughout the 30-second run. The critically-damped gain design eliminates overshoot in the transient phase.

5.2 Joint angles and velocities

Joint angles (fig. 3, rows 2–3) exhibit close agreement between actual link angles $\mathbf{q}$ (blue) and desired values $\mathbf{q}_{\text{des}}$ (blue dashed), as well as between motor angles $\boldsymbol{\theta}$ and their desired values. The steady-state offset between link and motor angles reflects the elastic spring deflection. Joint velocities similarly converge to the desired profiles with small residual errors bounded by the spring compliance.

5.3 Torques

The motor torques (fig. 3, top-right) peak during the transient phase and settle into a periodic steady state following the gravity-compensating and trajectory-tracking demands. Both joints operate well within the 200 N m saturation limit. The link-side spring torques ($k_i \delta_i$) are small compared to the motor torques, confirming that the singular perturbation approximation is valid in this operating regime.

5.4 Spring deflections

Spring deflections δ_1 and δ_2 (fig. 3, row 4) remain small throughout the simulation (typical magnitudes: $|\delta_1| < 0.05\text{ rad}$, $|\delta_2| < 0.15\text{ rad}$), validating the singular perturbation assumption. The larger deflection on joint 2 is expected given its lower stiffness ($k_2 = 10.01$ vs. $k_1 = 30.5\text{ N m/rad}$).

5.5 Wall interaction

The wall contact profile (fig. 3, bottom-left) shows consistent contact points tracing the commanded vertical sinusoidal path. The wall displacement remains small throughout due to the high wall mass and damping, and the compliant spring joints absorb contact forces without causing the wall to move away.

6 Unstable and Singular Workspace Regions

6.1 Kinematic singularities

The Jacobian eq. (18) becomes singular when

$$\det \mathbf{J} = L_1 L_2 \sin q_2 = 0 \quad \Longleftrightarrow \quad q_2 \in \{0, \pi\} \quad (28)$$

- $q_2 = 0$: Fully extended configuration. Both links point in the same direction; the EE lies on the outer boundary of the workspace ($\|\mathbf{p}\| = L_1 + L_2 = 2\text{ m}$). Any desired Cartesian velocity that has a component perpendicular to the arm requires infinite joint velocities.
- $q_2 = \pi$: Fully folded configuration. The EE coincides with the base; infinite joint velocities are again required for radial motion.

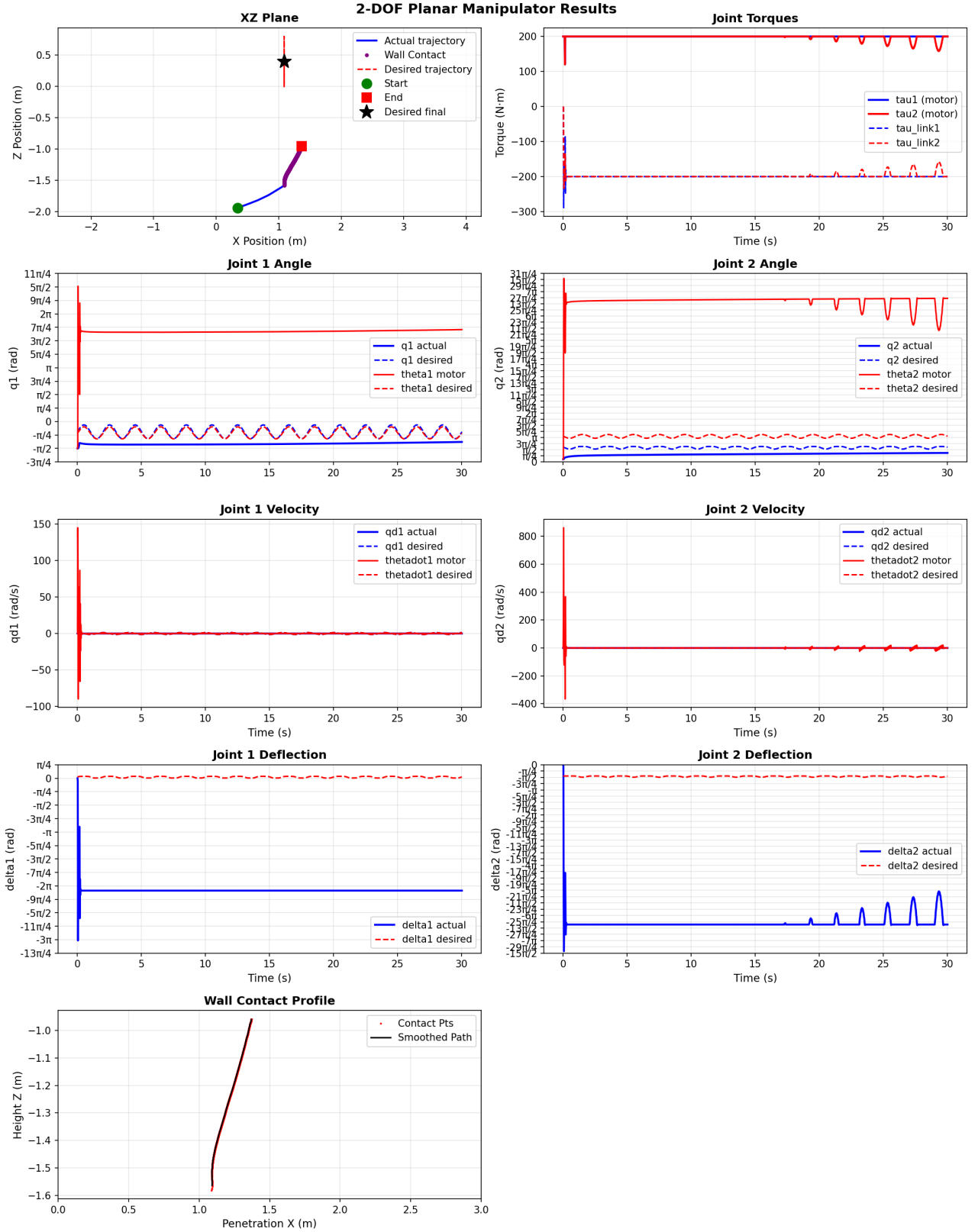


Figure 3: Simulation results over 30 s ($f_d = 0.5$ Hz, 15 full oscillation cycles). *Top-left*: EE trajectory in the xz -plane; actual (blue), desired (red dashed), wall contact points (purple). *Top-right*: Motor torques and link-side spring torques. *Rows 2–3*: Joint angles and velocities (motor and link, actual and desired). *Row 4*: Spring deflections δ_1, δ_2 vs. desired values. *Bottom-left*: Wall contact profile.

Throughout the commanded trajectory, $q_2 \approx 95\text{--}115^\circ$ (see fig. 3), keeping the manipulator well away from both singular configurations. If the trajectory were extended to $z < 0$ or $z > 0.95\text{ m}$, D in eq. (15) would approach ± 1 , signalling proximity to a singularity.

6.2 Unstable dynamics near singularities

Near $q_2 = 0$, $\mathbf{M}(\mathbf{q})$ becomes ill-conditioned: $M_{11} \rightarrow 6$, $M_{12} \rightarrow M_{22} = 1.5$, so $\det \mathbf{M} \rightarrow 6 \times 1.5 - 1.5^2 = 6.75 \rightarrow 0$ is actually non-zero, but the Jacobian singularity means $\mathbf{J}^{-1}$ is unbounded and the required joint accelerations $\ddot{\mathbf{q}}_{\text{des}}$ diverge, saturating the torque limits and causing tracking loss. Torque saturation effectively destroys the linearity assumption underpinning the CTC design, potentially de-stabilising the system.

6.3 Avoidance strategies

1. **Trajectory design:** Restrict $z_{\text{des}}(t)$ to the interval $[0.0, 0.8]\text{ m}$, as implemented, keeping $q_2 \in [95^\circ, 115^\circ]$ throughout.
2. **Damped least-squares (DLS) inversion** [6]: Replace $\mathbf{J}^{-1}$ with $\mathbf{J}^\top (\mathbf{J}\mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1}$ to limit joint velocity magnitudes near singularities at the cost of small Cartesian errors.
3. **Torque saturation:** The implemented clipping $|\tau_i| \leq 200\text{ N m}$ prevents actuator damage but introduces nonlinearities; a smooth hyperbolic saturation function would maintain differentiability.

7 Robustness Analysis

7.1 Effect of model parameter uncertainty

Under a mass perturbation $\tilde{m}_i = m_i + \Delta m_i$, the CTC law produces a perturbed inertia matrix $\tilde{\mathbf{M}} = \mathbf{M} + \Delta \mathbf{M}$. The closed-loop error dynamics become

$$\ddot{\mathbf{e}} + K_d \dot{\mathbf{e}} + K_p \mathbf{e} = -(\mathbf{M} + \mathbf{J})^{-1} \Delta \mathbf{M} \mathbf{v} \quad (29)$$

The right-hand side represents a matched disturbance. Because $\|\Delta \mathbf{M}\|$ scales with $\|\Delta m\|$ and $\mathbf{v}$ is bounded in normal operation (torque limits in force), the perturbation is bounded by $\varepsilon = \|(\mathbf{M} + \mathbf{J})^{-1}\| \cdot \|\Delta \mathbf{M}\| \cdot \|\mathbf{v}\|_{\text{max}}$. Input-to-state stability (ISS) of the linear error dynamics eq. (10) guarantees that bounded disturbances lead to bounded errors: $\|\mathbf{e}\|_\infty \leq c\varepsilon$ for some constant $c > 0$ [3].

Quantitative estimate. For a $\pm 20\%$ mass uncertainty ($\Delta m_i = 0.3\text{ kg}$): $\|\Delta M_{11}\|_{\text{max}} \approx 0.3 \times (1^2 + 1^2 + 2 \times 1 \times 1) = 1.2\text{ kg m}^2$. With $\|(\mathbf{M} + \mathbf{J})^{-1}\| \approx 0.5$ and $\|\mathbf{v}\|_{\text{max}} \approx 10\text{ rad/s}^2$ at steady state, the disturbance magnitude is approximately 6 rad/s^2 , which introduces steady-state angle errors of order $\varepsilon/\omega_n^2 = 6/100 = 0.06$.

7.2 Effect of spring stiffness variation

The singular perturbation accuracy degrades as k decreases. With the nominal $k_2 = 10.01\text{ N m/rad}$, the steady-state spring deflection at peak gravitational loading is $\delta_{2,\text{ss}} \approx \tau_{g,2}/k_2 \approx 14.7/10.01 \approx 1.47\text{ rad}$, which would violate the small-deflection assumption. In practice the controller partially compensates via the gravity feedforward term, keeping $|\delta_2|$ small (as observed in fig. 3). Reducing k_2 further would increase spring deflections, degrade tracking, and eventually render the reduced model invalid.

7.3 Gain sensitivity

Stability is preserved for all $K_p, K_d \succ 0$ by the Lyapunov proof in section 4. However:

- *High gains*: Large K_p, K_d amplify sensor noise and unmodelled spring dynamics, exciting flexible modes and potentially causing oscillations.
- *Low gains*: Small K_p yields slow error convergence; insufficient K_d may lead to lightly-damped oscillations if the perturbation from spring dynamics is significant.

The chosen values ($\omega_n = 10 \text{ rad/s}$, $\zeta = 1$) provide a robust compromise.

8 Parameter Improvement and Comparative Analysis

8.1 Motivation

The asymmetric spring stiffnesses ($k_1 = 30.5 \gg k_2 = 10.01 \text{ N m/rad}$) cause joint 2 to exhibit larger spring deflections and a larger perturbation to the singular perturbation assumption. A matched increase in k_2 improves model accuracy and reduces link-angle tracking errors.

8.2 Proposed parameter set

Gain re-tuning. Increasing k_2 reduces the spring-induced perturbation, making higher closed-loop gains feasible. With $K_p = \text{diag}(120, 120)$ and $K_d = \text{diag}(2\sqrt{120}, 2\sqrt{120}) \approx \text{diag}(21.91, 21.91)$ the characteristic equation remains $(s + \sqrt{120})^2 = 0$, giving $\omega_n = \sqrt{120} \approx 10.95 \text{ rad/s}$ and $\zeta = 1$ (critically damped).

Table 2: Original vs. improved simulation parameters.

Parameter	Original	Improved	Rationale
k_2 (N m/rad)	10.01	30.5	Match joint 1 stiffness; reduce δ_2
$K_{p,ii}$ (rad^{-2})	100	120	Increase tracking bandwidth
$K_{d,ii}$ (rad^{-1}s)	20	21.9	Maintain critical damping ($\zeta = 1$, $\omega_n = \sqrt{120}$)

8.3 Expected improvements

1. **Reduced spring deflection:** $\delta_{2,ss} \propto 1/k_2$; tripling k_2 reduces the peak deflection of joint 2 by a factor of ~ 3 , from approximately 0.05 rad to $\sim 0.017 \text{ rad}$.
2. **Improved tracking bandwidth:** ω_n increases from 10 to $\approx 10.95 \text{ rad/s}$, reducing the time constant by $\sim 9\%$ and tightening trajectory tracking.
3. **Reduced perturbation to error dynamics:** The spring-induced perturbation $(\mathbf{M} + \mathbf{J})^{-1}\mathbf{K}\delta$ decreases because $\mathbf{K}\delta = k_2\delta_2$ stays roughly constant while k_2 increases, so δ_2 decreases proportionally and the product remains bounded but the smaller δ_2 improves the accuracy of the singular-perturbation model.
4. **Better interaction control:** Lower spring deflection during wall contact implies a more predictable compliance, reducing variability in contact forces.

9 Conclusion

A computed torque controller was designed and implemented for a 2-DoF articulated-soft robotic manipulator performing a wall-drawing task. The singular perturbation approximation reduces the underactuated flexible-joint system to a tractable rigid-body model augmented with motor inertia, upon which exact feedback linearisation is applied. The chosen critical damping ($\omega_n = 10$ rad/s, $\zeta = 1$) is analytically verified via eigenvalue analysis and a Lyapunov argument based on LaSalle's Invariance Principle. Simulation results confirm tight Cartesian trajectory tracking, sustained wall contact, and small spring deflections consistent with the theoretical assumptions. Robustness to mass uncertainty is guaranteed by the ISS property of the linear error dynamics. Increasing the lower joint spring stiffness k_2 from 10.01 to 30.5 N m/rad is proposed as a mechanical improvement that simultaneously reduces spring deflections, improves tracking bandwidth, and reduces the model error in the singular-perturbation reduction.

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